

"Raspberry Pi enables embedded engineering while being a cheap, low-power Linux platform"

Raspberry Pi has transformed the world of computing. This single-board computer has opened new avenues for millions of budding developers all across the globe. But what was the prime idea behind the tiny little device and how did it receive such worldwide acclaim?

Jagmeet Singh of OSFY tried to get the answers to these questions in an exclusive conversation with **Raspberry Pi** creator, **Eben Upton**. Read on for the edited excerpts.

Q How did you get the idea to design Raspberry Pi?

When I was a child, I had two different small computers — BBC Micro and Amiga. At the time that I went to university and started computing, both the computers were quite common for people of my age in the UK. So to engage a hobbyist's interest and to step into the computer industry, I planned to develop Raspberry Pi.

Raspberry Pi filled the space that was left empty by machines like BBC Micro and Amiga. It was really an attempt to provide kids with the same experience that we used to have back in the 1980s.

Q What is the real mission behind Raspberry Pi? Is it just to give the world an affordable computing experience or also to open new avenues for Internet of Things (IoT) developments?

The mission is quite simple and it is to give young people access to a computer science education experience. We didn't design Raspberry Pi for the industrial control space, but it turns out to be very well designed for industrial control and the IoT world.

We just want to create a new generation of people who are excited about computing and computer programming. Mainly, we are targeting people who are not just the end users of computers but are also potential creators. They are the users who understand computing at a very deep level.

Q Raspberry Pi is already helping students learn new code. How can it be an impactful device in the education sector?

Obviously, several factors make Raspberry Pi an impactful device in the education sector. One is affordability, as this is incredibly important everywhere. There is a misconception about computing, which is that everybody has a computer. But most of the users all across the globe were using one family computer in the past, due to its high price. In addition,

a significant percentage of people had devices with some closed platforms. Thus, they were scared to break the devices and even avoided customising them using their own code and tweaks.

We were determined to generate a solution that would be cheap enough, and Raspberry Pi emerged as the result. One of the major things that persuade you to buy a Raspberry Pi is that if you break the board, you can certainly afford a new one.

The other thing that makes Raspberry Pi a perfect solution for students is the feasible computing experience. We have provided the universally accepted 40-pin GPIO header that helps children try some of their advanced sensors and transform some simple projects into intuitive ones. Kids use

Raspberry Pi to write new software and control things in the real world.

Q What are your plans to expand the presence of Raspberry Pi in the Indian market?

We are keen to get the cost down and availability up for Raspberry Pi in India, and want to reduce the delay in introducing new products in the Indian market. We are also looking at some initiatives with the state governments in the country.

On the embedded systems side, we are trying to improve the cost. There are plans to work with various government bodies in the country to fulfil the needs of the educational sector. We want to see Raspberry Pi in the hands of a large number of children in India who are eager to learn computing. We also want to make sure that there is appropriate education and support material in the Indian market to help those children.

Q The Indian government has recently launched some initiatives like 'Digital India' and 'Skill India'. Do you think it would be worth it for the Raspberry Pi Foundation to participate in both these programmes?

Because we are a very small organisation, and it takes some major effort to join some flagship local government programme, we are unlikely to get involved directly.

